$$f\left(k\right) = \left(1 - e^{-\frac{k \cdot n}{m}}\right)^k$$

Note that `n` and `m` are constants in the formula above.

We will go into the details of finding the minimum for `f(k)` in the next section; for now (or if you are more interested in the result), just know that the optimal value is:

$$k^* = \frac{m}{n} \cdot ln(2)$$

Now that we have a formula for k^*, we can substitute it in our previous formula, the one for `p(n, m, k)`, and after a some algebraic manipulation, we obtain an expression for the optimal value for `m` (let's call it m^*):

$$m^* = -n \cdot \frac{ln(p)}{ln(2)^2}$$

This means that, if we know in advance the total number `n` of unique elements that we will insert in the container, and we set the probability `p` of a false positive to the largest value that is acceptable, then we can compute the optimal size of the buffer (and the number of bits per key we need to use) in order to guarantee the desired precision.

Two aspects are significant, when looking at the formulas we derived: